Wave motion (Supplementary Problems)

1. Figure 40.1 illustrates a thin film made of a transparent material with a thickness d and refractive index n_f, where $n_2 > n_f > n_1$. What are the three smallest thicknesses for which the reflected light rays 1 and 2 will interfere (a) constructively and (b) destructively? Assume the wavelength of light in the film is 600 nm.

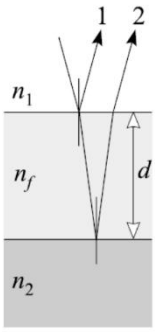


Fig. 40-1

2. Two narrow horizontal slits, separated by a distance of a = 0.60 mm, are illuminated by a beam of 500-nm light, as shown in Fig. 40-2. Light diffracted at specific angles θ reinforces at some angles and cancels at others. Determine the three smallest values of θ for which (a) reinforcement occurs and (b) cancellation occurs. (Refer to Fig. 40-3.)

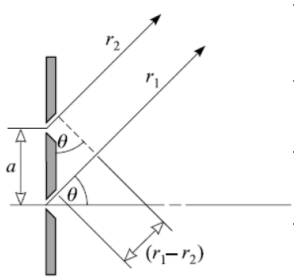


Fig. 40-2

3. Monochromatic light from a point source illuminates two narrow, horizontal parallel slits, with their centers $a = 0.80$ mm apart, as shown in Fig. 40-3. An interference pattern forms on a screen located 50 cm away. In this pattern, the bright and dark fringes are evenly spaced. The distance y indicated is 0.304 mm. Calculate the wavelength λ of the light.

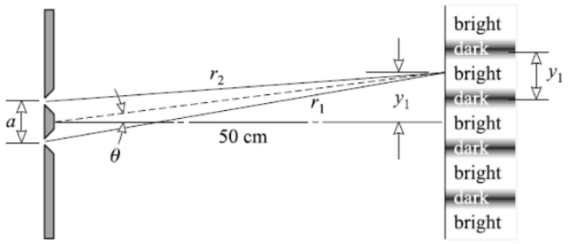


Fig. 40-3

4. A double-slit experiment is conducted with 480-nm light and narrow slits that are 0.050 cm apart. At what angle to the straight-through beam will one observe (a) the third-order bright spot and (b) the second minimum from the central maximum?

5. In Problem 4, if the distance from the slit to the screen is 200 cm, how far from the central maximum are (a) the third-order bright spot and (b) the second minimum?

6. Red light with a wavelength of 644 nm passes through two parallel narrow slits that are 1.00 mm apart. Determine the distance between the central bright fringe and the third dark interference fringe formed on a screen parallel to the plane of the slits and 1.00 m away.

7. Two flat glass plates are pressed together at the top edge and separated at the bottom edge by a strip of tinfoil. The air wedge is examined using yellow sodium light (589 nm) reflected normally from both surfaces, resulting in the observation of 42 dark interference fringes. Calculate the thickness of the tinfoil.

8. A mixture of yellow light with a wavelength of 580 nm and blue light with a wavelength of 450 nm is incident normally on an air film 290 nm thick. What color is the reflected light?

9. A single slit with a width of 0.140 mm is illuminated by monochromatic light, creating diffraction bands on a screen 2.00 m away. If the second dark band is located 16.0 mm from the central bright band, what is the wavelength of the light?

10. A narrow beam of yellow light with a wavelength of 600 nm is incident normally on a diffraction grating ruled with 2000 lines/cm. Images are formed on a screen parallel to the grating and 1.00 m away. Calculate the distance along the screen from the central bright line to the first-order lines.

11. The distance between the two slits is 0.030 mm. The second-order bright fringe (m= 2) is measured on a viewing screen at an angle of 2.15° from the central maximum. Determine the wavelength of the light.

12. A double-slit interference experiment is performed with blue-green light from an argon-gas laser. The separation between the slits is 0.50 mm, and the first-order maximum of the interference pattern is at an angle of 0.059° from the center of the pattern. What is the wavelength of argon laser light?

13. Light falls on a double slit with slit separation of 2.02×10^{-6} m, and the first bright fringe is seen at an angle of 16.5° relative to the central maximum. Find the wavelength of the light.

14. A pair of narrow parallel slits separated by a distance of 0.250 mm is illuminated by the green component from a mercury vapor lamp ($\lambda = 546.1$ nm). Calculate the angle from the central maximum to the first bright fringe on either side of the central maximum.

15. Monochromatic light from a helium-neon laser ($\lambda = 632.8$ nm) shines at a right angle to the surface of a diffraction grating that contains 150 500 lines/m. Find the angles at which one would observe the first order and second-order maxima.

16. A diffraction grating with 5.000×10^3 lines/cm is used to examine the sodium spectrum. Calculate the angular separation of the two closely spaced yellow lines of sodium (588.995 nm and 589.592 nm) in each of the first three orders.

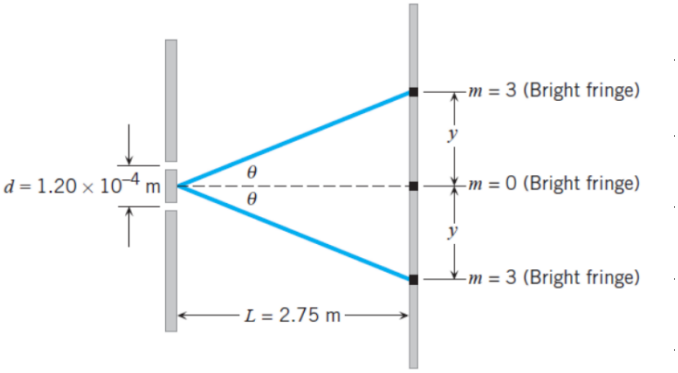
17. A diffraction grating with 4525 lines/cm is illuminated by direct sunlight. The first-order solar spectrum is spread out on a white screen hanging on a wall opposite the grating.

a. At what angle does the first-order maximum for blue light with a wavelength of 422 nm appear?

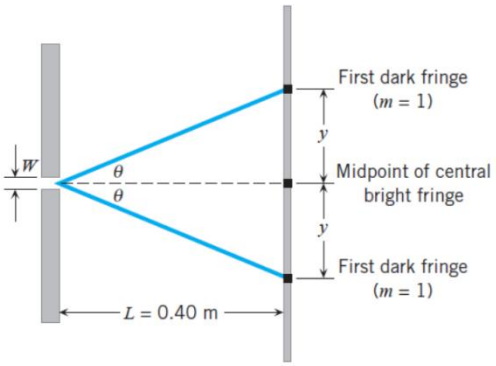
b. At what angle does the first-order maximum for red light with a wavelength of 655 nm appear?

18. A diffraction grating is calibrated by using the 546.1 nm line of mercury vapor. The first-order maximum is found at an angle of 21.2°. Calculate the number of lines per centimeter on this grating.

19. Red light ($\lambda = 664$ nm in vacuum) is used in Young's experiment, with the slits separated by a distance of $d = 1.20 \times 10^{-4}$ m. The screen shown in Figure 27.7 is positioned 2.75 m from the slits. Determine the distance y on the screen between the central bright fringe and the third-order bright fringe.



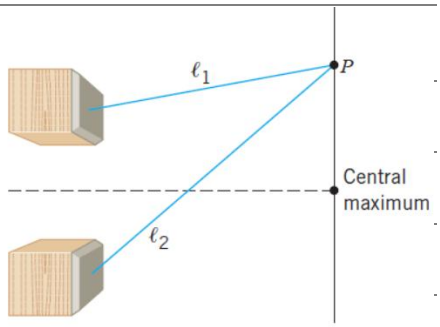
20. Light passes through a slit and shines on a flat screen located $L = 0.40$ away (see Figure 27.22). The wavelength of the light in a vacuum is $\lambda = 410$ nm. The distance between the midpoint of the central bright fringe and the first dark fringe is y. Determine the width $2y$ of the central bright fringe for the following slit widths: (a) $W = 5.0 \times 10^{-6}$ m and (b) $W = 2.5 \times 10^{-6}$ m.



21. A mixture of violet light ($\lambda = 410$ nm in vacuum) and red light ($\lambda = 660$ nm in vacuum) is directed onto a grating that contains $1.0 \times 1\,0^{4}$ lines/cm. For each wavelength, determine the angle θ that locates the first-order maxima.

22. The two loudspeakers in the drawing are producing identical sound waves. The waves spread out and overlap at the point *P*. What is the difference $L_2 - L_1$ in the two path lengths if point P is at the third sound intensity minimum from the central sound intensity maximum? Express this difference in terms of the wavelength λ of the sound.

a. ½ λ b. λ c. 3λ d. 5/2λ



23. A single slit has a width of 2.1×10^{6} m and is used to form a diffraction pattern. Find the angle that locates the second dark fringe when the wavelength of the light is **(a)** 430 nm and **(b)** 660 nm.

24. A diffraction pattern is created when light passes through a single slit. The wavelength of the light is 675 nm. Determine the angle that locates the first dark fringe for the following slit widths: (a) $1.8 \times 1\,0^{-4}$ m and (b) $1.8 \times 1\,0^{-6}$ m.

25. A flat screen is positioned 0.60 m away from a single slit. Light with a wavelength of 510 nm (in vacuum) passes through the slit and creates a diffraction pattern. The width of the central bright fringe on the screen measures 0.050 m. What is the width of the slit?

26. Light passes through a single slit with a width of $5.6 \times 1\,0^{-4}$ m. A diffraction pattern is formed on a flat screen located 4.0 m away. The distance between the center of the central bright fringe and the first dark fringe is 3.5 mm. What is the wavelength of the light?

27. The width of a slit is $2.0 \times 1\,0^{-5}$ m. Light with a wavelength of 480 nm passes through this slit and strikes a screen located 0.50 m away. In the resulting diffraction pattern, determine the width of the bright fringe adjacent to the central bright fringe.

28. Light waves with two different wavelengths, 632 nm and 474 nm, pass simultaneously through a single slit with a width of $7.15 \times 1\,0^{-5}$ m and strike a screen 1.20 m from the slit. Two diffraction patterns are formed on the screen. What is the distance (in cm) between the common center of the diffraction patterns and the first occurrence of a dark fringe from one pattern coinciding with a dark fringe from the other pattern?

29. In a Young's double-slit experiment, the wavelength of the light used is 520 nm (in vacuum), and the separation between the slits is $1.4 \times 1\,0^{-6}$ m. Determine the angle that locates: (a) the dark fringe for which $m = 0$, (b) the bright fringe for which $m = 1$ (c) the dark fringe for which $m = 1$ (d) the bright fringe for which $m = 2$

30. The dark fringe for $m = 0$ in a Young's double-slit experiment is located at an angle of $\theta = 1\,5^{\circ}$. What is the angle that locates the dark fringe for $m = 1$?

31. A flat observation screen is positioned 4.5 m away from a pair of slits. The separation on the screen between the central bright fringe and the first-order bright fringe is 0.037 m. The light illuminating the slits has a wavelength of 490 nm. Determine the slit separation.

32. Light with a wavelength of 668 nm passes through a slit that is $6.73 \times 1\,0^{-6}$ m wide and falls on a screen located 1.85 m away. What is the distance on the screen from the center of the central bright fringe to the third dark fringe on either side?